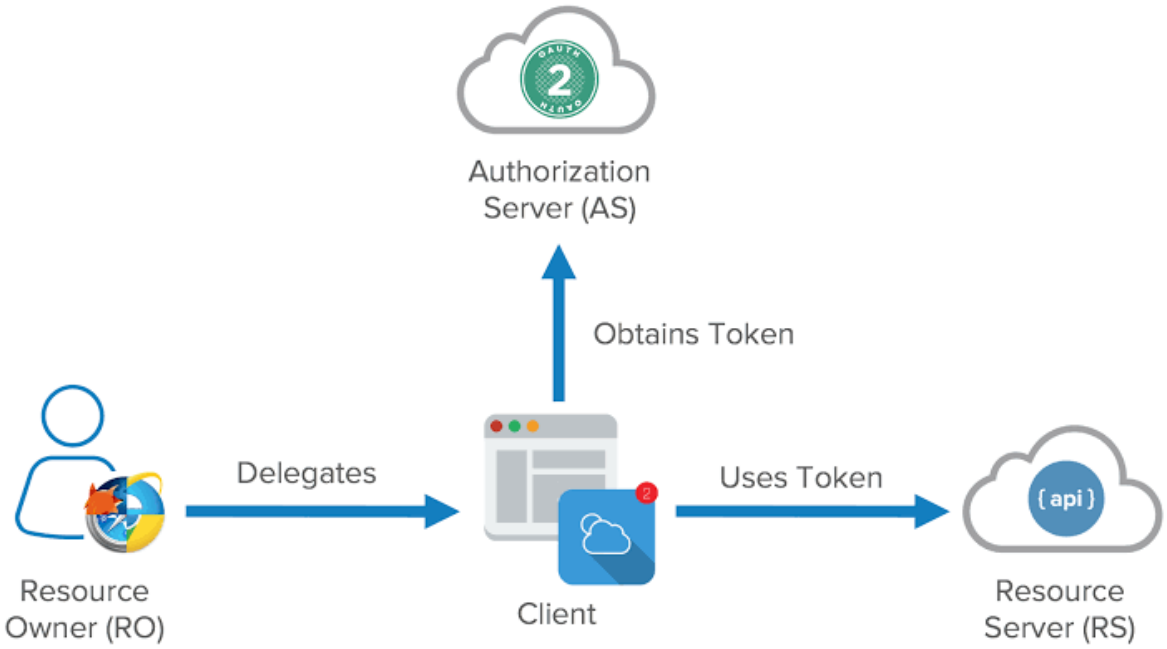
Security, Performance & Code Quality

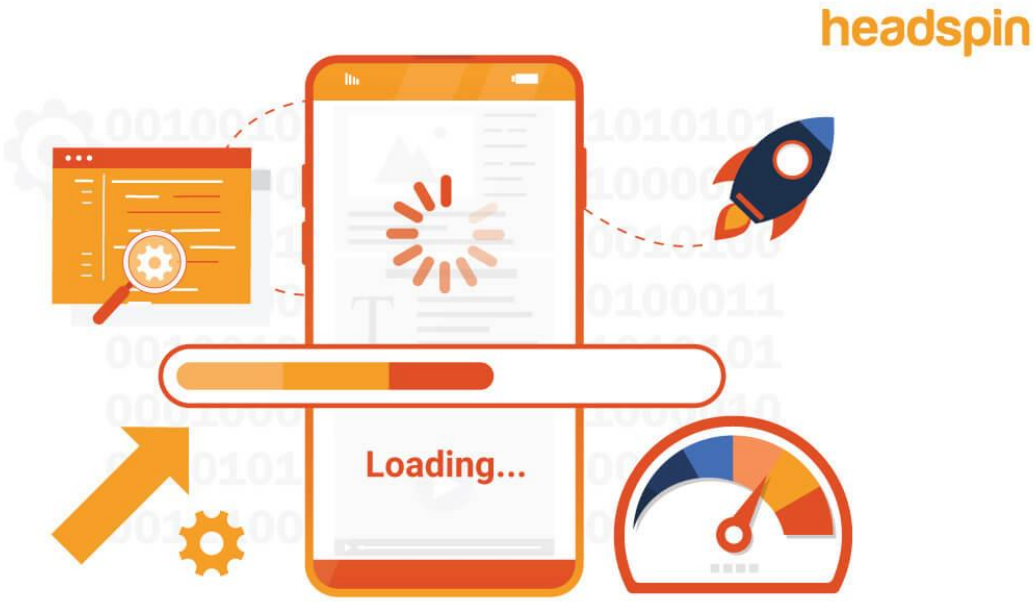
Security Compass

THE SECURITY BY DESIGN COMPANY

Secure Software Development Life Cycle (Secure SDLC)

- Planning Phase
- Design Phase
- Implementation Phase
- Verification Phase
- Maintenance Phase





Purpose of Security, Performance & Code Quality

A software application is only successful if it is:

- **Secure** (protects users & data)
- **Fast** (performs well under load)

- **Maintainable** (easy to understand and improve)

At **ENGLORAY Tech Group**, this resource ensures developers build **reliable, scalable, and professional-grade software**, not fragile systems.

1. Application Security (Protecting Users & Data)

Security is built **from the first line of code**, not added later.

◇ Core Security Practices

- Authentication & authorization
- Role-based access control
- Password hashing & encryption
- Secure API endpoints
- Input validation & sanitization

◇ Common Security Threats Prevented

- SQL Injection
- Cross-Site Scripting (XSS)
- Cross-Site Request Forgery (CSRF)
- Unauthorized access

Tools & Standards Used

- Token-based authentication (JWT)
- HTTPS & SSL
- Secure environment variables

Outcome: Applications users can trust with their data.

2. Performance Optimization (Speed & Scalability)

Performance directly affects **user experience and business success**.

◇ Performance Techniques Used

- Efficient database queries
- Caching strategies
- Asynchronous processing
- Load handling & optimization
- API response optimization

◇ Performance Monitoring

- Response time analysis
- Error tracking
- Server resource monitoring

Why ENGLORAY Emphasizes Performance

✓ Faster load times
✓ Higher user retention
✓ Better SEO & system reliability

3. Clean Code & Maintainability (Code Quality)

Clean code is code that is **easy to read, understand, and modify**.

◇ Code Quality Principles

- Meaningful variable & function names
- Modular & reusable components
- Proper folder structure
- Clear comments where needed

◇ Standards Followed

- SOLID principles
- DRY (Don’t Repeat Yourself)

- Consistent formatting

Tools Used

- **ESLint**
- **Prettier**

Outcome: Code that teams can maintain long-term without confusion.

4. Testing & Bug Prevention

Testing ensures software works **before users face problems**.

◇ Testing Types

- Unit testing
- Integration testing
- API testing
- Regression testing

Tools Used

- **Postman**
- Framework-based testing tools

Why Testing Matters

✓ Fewer bugs in production
✓ Safer updates
✓ Higher software reliability

5. Code Reviews & Team Standards

At ENGLORAY, code is **reviewed, not rushed**.

Code Review Focus

- Security risks
- Performance issues
- Code readability
- Architecture consistency

Benefits of Code Reviews

✓ Knowledge sharing
✓ Fewer mistakes
✓ Higher overall code quality

6. Continuous Improvement & Best Practices

Security, performance, and quality are **ongoing processes**.

ENGLORAY Approach

- Regular refactoring
- Updating dependencies
- Monitoring real-world usage
- Improving based on feedback

Final Outcome of Security, Performance & Code Quality

With this resource, ENGLORAY software developers can:

✓ Build secure applications
✓ Deliver fast, optimized systems
✓ Write clean, professional code
✓ Reduce bugs & technical debt
✓ Create long-lasting software products